

Binary to Decimal Conversion

Others to Decimal:

$$(a_3 a_2 a_1 a_0 . a_{-1} a_{-2})_r \longrightarrow (\quad)_{10}$$

$$= \boxed{a_3 r^3 + a_2 r^2 + a_1 r^1 + a_0 r^0 + a_{-1} r^{-1} + a_{-2} r^{-2} = (\quad)_{10}}$$

Ex: $(\underset{\substack{4 \ 3 \ 2 \ 1 \ 0 \ -1 \ -2}}{1 \ 0 \ 1 \ 0 \ 1 . 1 \ 1})_2 \longrightarrow (\quad)_{10}$

Binary print

$$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$a_0 = 1 \quad a_{-1} = 1 \quad r = 2$$

$$a_1 = 0$$

$$a_{-2} = 1$$

$$a_2 = 1$$

$$a_3 = 0$$

$$a_4 = 1$$

$$16 + 0 + 4 + 0 + 1 + \frac{1}{2} + \frac{1}{4}$$

$$21 + \frac{2+1}{4}$$



Ex: $(10101.11)_2 \longrightarrow (21.75)_{10}$

$\begin{matrix} 4 & 3 & 2 & 1 & 0 & -1 & -2 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{matrix}$

Binary point

$$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$16 + 0 + 4 + 0 + 1 + \frac{1}{2} + \frac{1}{4}$$

$$21 + \frac{2+1}{4}$$

$$21 + \frac{3}{4}$$

$$21 + 0.75 = (21.75)_{10}$$

$$\begin{aligned} a_0 &= 1 & a_{-1} &= 1 & r &= 2 \\ a_1 &= 0 & a_{-2} &= 1 \\ a_2 &= 1 \\ a_3 &= 0 \\ a_4 &= 1 \end{aligned}$$



Ex.

$$(1101100.101)_2 \rightarrow ()_{10}$$

108.625